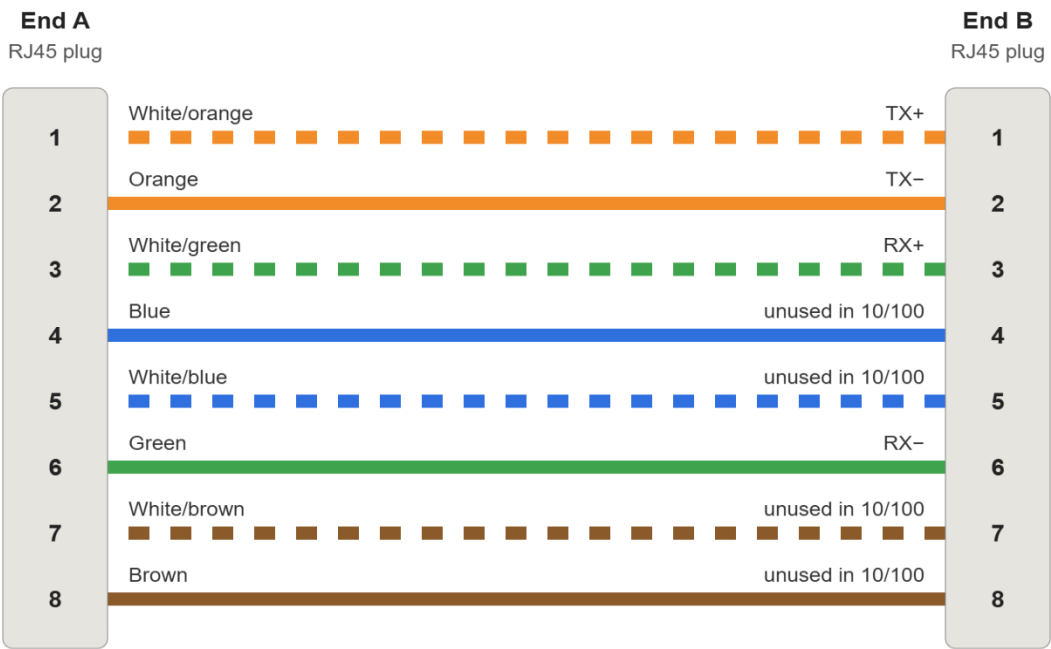


# Network Cable Types and ToolsTheory

1. List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

| Feature             | Ethernet Cable                                                                                                                                         | Fiber Optic Cable                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Transmission Medium | Uses <b>copper wires</b> to transmit electrical signals.                                                                                               | Uses <b>glass or plastic fibers</b> to transmit data using light.                                                                            |
| Speed & Distance    | Suitable for high-speed connections but generally has a shorter maximum distance, commonly up to <b>100 meters</b> for standard Ethernet copper links. | Supports <b>very high speeds</b> and much longer distances, from hundreds of meters to many kilometers depending on the fiber and equipment. |
| Interference        | Can be affected by <b>electromagnetic interference (EMI)</b> .                                                                                         | Resistant to electromagnetic interference because it uses <b>light instead of electrical signals</b> .                                       |

2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



| Pin | Wire Color   | Function         |
|-----|--------------|------------------|
| 1   | White/Orange | TX+              |
| 2   | Orange       | TX-              |
| 3   | White/Green  | RX+              |
| 4   | Blue         | Unused in 10/100 |
| 5   | White/Blue   | Unused in 10/100 |
| 6   | Green        | RX-              |
| 7   | White/Brown  | Unused in 10/100 |
| 8   | Brown        | Unused in 10/100 |

#### Wiring

order:

**White/Orange → Orange → White/Green → Blue → White/Blue → Green → White/Brown → Brown**

End

A:

T568B

End B: T568B

3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

<https://youtu.be/N1dcCzfgdhs?si=lpl-zZF8KDrbc1c>

#### Steps to Attach an RJ45 Connector to a CAT6 Cable

1. First, cut the CAT6 cable to the required length and put the **RJ45 boot** on the cable if you are using one.
2. Remove a small part of the outer covering of the cable to expose the internal wires.
3. Separate and untwist all the wires.
4. Arrange the wires according to the **T568B standard: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.**
5. Make all the wires straight and keep them flat in the correct order.
6. Cut the wire ends evenly so they can properly fit inside the RJ45 connector.
7. Insert all 8 wires into the RJ45 connector and make sure they reach the end.
8. Check once again that the wire order is correct and the cable jacket is inside the connector.
9. Put the connector into the **RJ45 crimping tool** and squeeze it firmly.

10. Finally, check that the connector is properly attached. If available, use a **cable tester** to check the connection.

4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences

For a gaming café with **10 PCs**, I would use **Ethernet cables (CAT6)** because they are affordable, easy to install, and suitable for LAN gaming. I would use an **RJ45 crimper** to attach RJ45 connectors to the CAT6 cables. Ethernet cables can provide reliable and fast connections between the PCs and the network switch. **Fiber optic cables and a fiber cleaver** would be more suitable for long-distance or backbone connections rather than connecting individual PCs in a small café.